

XShooter data reduction README

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For P105.20KF.001 and P108.22JL.001

0101.A-0577 was reduced in a similar way by Rebecca Allen (see that specific readme for details)

Directory structure:

<Object name (as used in the ESO observations)>

<deflector or source>

reduce.sh

<intermediary>

<date of observations>

<timestamp of observations UVB >/reduced 2D UVB file.fits

<timestamp of observations NIR >/reduced 2D NIR file.fits

<timestamp of observations VIS >/reduced 2D VIS file.fits

<telluric files>/ telluric files for all three arms

<reduced>

<data of observation>

1D extracted UVB, NIR, VIS files for the for the blobs A and B as well as the profile showing the extraction. The 2D files are similar to those in intermediary but show regions where skylines etc were masked.

<blobA (and blobB if multiple blobs)>

Binned ($_bx$ where x is the number of binned pixels) and smoothed ($_sx$ in a gaussian with width x pixels). There are also unbinned and unsmoothed versions of the spectra here without a suffix. Stacked 2D spectra are noted with $_spec2d$ and its residuals are $_spec2d_sub$.

Step 1: 2D spectra with ESOReflex

All the data was first reduced using the esoreflex software using the default setting for the XShooter pipeline. See [Reflex X-Shooter Tutorial](#) for full details. The 2D spectra are flux calibrated, and are in air wavelengths (see [XShooter FAQ](#)). They are not corrected for telluric absorption. The flux units are in the fits headers (erg/s/cm²/Angstrom). The reduced 2D spectra output by esoreflex are in the following directories:

<objname>/<deflector or source>/intermediary/<date of observation>/<uv, optical, or near infrared arm>/.

Step 2: optimised 1D extractions with using [speclib](#)

The ESOReflex pipeline does make 1D extractions but they aren't very good, so I instead used code written Corentin Schreiber and available on his github that was specifically designed to do improved XShooter 1D extractions.

To do these improved extractions involved visually inspecting the 2D spectra and identifying (a) how many blobs there are and (b) where the blobs are in the slit (an initial guess to provide the code). This information was then written into a reduce.sh file (which exists in each <objname>/<deflector or source>/ directory).

In the reduce.sh file I also specify various binning and smoothing schemes for the output 1D spectra. Currently the data is spectrally binned (_b) and spectrally smoothed in a gaussian window (_s) within windows of 10 and 50 pixels.

More details on how Corentin's speclib code works can be read [here](#).

These 1D spectra are flux calibrated and are in air wavelengths (see [XShooter FAQ](#)). The flux units are in erg/s/cm²/Angstrom.